ವಿಶ್ವೇಶ್ವರಯ್ಯ ತಾಂತ್ರಿಕ ವಿಶ್ವವಿದ್ಯಾಲಯ, ಬೆಳಗಾವಿ

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

KARNATAKA, INDIA

Certifies that

RAHUL HEGDE

ಬ್ಯಾಚಲರ್ ಆಫ್ ಇಂಜಿನಿಯರಿಂಗ್

ಪದವಿಗೆ ಅಗತ್ಯವಾದ ಅರ್ಹತೆಗಳನ್ನುಳ್ಳವರೆಂದು ಪರಿಗಣಿತವಾಗಿರುವ ಕಾರಣ ಮೇಲೆ ಕಾಣಿಸಿದ ಪದವಿಗೆ ಅವರನ್ನು ಅಂಗೀಕರಿಸಲಾಗಿದೆ.

has been duly admitted to the Degree of

Bachelor of Engineering

in recognition of the fulfilment of requirements

for the said degree through

ವಿದ್ಯಾಲಯದ ಹೆಸರು Name of the College	:	*Nitte Mahalinga Adyanthaya Memorial Institute of Technology, Nitte (Autonomous College)*
ಪರೀಕ್ಷಾ ಕ್ರಮ ಸಂಖ್ಯೆ University Seat Number	:	**4NM16EC101**
ವಿಷಯ / Subject	:	*Electronics & Communication Engineering*
ಶ್ರೇಣಿ Class	:	*First Class with Distinction*

A 21 06176

ವಿಶ್ವವಿದ್ಯಾಲಯದ ಮುದ್ರೆಯೊಂದಿಗೆ ಕೊಡಲ್ಪಟ್ಟಿದೆ
Given under the seal of the University

ಬೆಳಗಾವಿ
Belagavi

ದಿನಾಂಕ
Date : April 03, 2021

ಕುಲಪತಿ
VICE CHANCELLOR

NITTE EDUCATION TRUST | NMAM INSTITUTE OF TECHNOLOGY

(An Autonomous Institution affiliated to VTU, Belagavi)

NMAM/ACA/2020/3380 07 – 09 - 2020

TRANSCRIPT AS PER RECORDS **CREDIT BASED SYSTEM AND GPA SCHEME OF EVALUATION**

Name: **RAHUL HEGDE**

Father's Name: **MOHAN HEGDE**

University Seat Number: **4NM16EC101**
Year of Entry: **2016**

Degree Programme Registered For:
BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATION ENGINEERING

1. Duration of Course: **4 Years**
2. Medium of Instruction: **ENGLISH**

Grade Points Scale for Grading

Grade	S	A	B	C	D	E	F
Grade Points	10	09	08	07	05	04	00
Score (Marks Range (%))	≥ 90	≥ 75 < 90	≥ 60 < 75	≥ 50 < 60	≥ 45 < 50	≥ 40 < 45	< 40

Sl. No	Subjects	Credits	Grade Point	Grade Letter	Month & Year of Passing
I SEMESTER					
1	ENGINEERING MATHEMATICS - I	4	10	S	Dec. 2016 – Jan. 2017
2	ENGINEERING PHYSICS	4	10	S	
3	ELEMENTS OF CIVIL ENGG. & ENGG. MECHANICS	4	9	A	
4	ELEMENTS OF MECHANICAL ENGG	4	10	S	
5	BASIC ELECTRICAL ENGG.	4	10	S	
6	ENGINEERING SKILL DEVELOPMENT LAB.	1	9	A	
7	CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS	1	10	S	
8	ENGINEERING PHYSICS LAB.	2	10	S	
9	KANNADA	0	-	PP	
10	BIOLOGY FOR ENGINEERS	0	-	PP	
SGPA: 9.79	**CGPA: ---**				
II SEMESTER					
1	ENGINEERING MATHEMATICS - II	4	10	S	May - June 2017
2	ENGINEERING CHEMISTRY	4	9	A	
3	COMPUTER CONCEPTS & 'C' PROGRAMMING	4	9	A	
4	BASIC ELECTRONICS	4	10	S	
5	ENVIRONMENTAL STUDIES	1	10	S	
6	ENGLISH & COMMUNICATION SKILLS	2	9	A	
7	COMPUTER AIDED ENGG. GRAPHICS	3	9	A	
8	COMPUTER CONCEPTS & 'C' PROGRAMMING LAB	2	10	S	
9	ENGINEERING CHEMISTRY LAB	2	10	S	
SGPA: 9.50	**CGPA: 9.64**				
III SEMESTER					
1	VECTOR CALCULUS AND TRANSFORM TECHNIQUES	4	9	A	Dec. 2017 – Jan. 2018
2	ANALOG ELECTRONIC CIRCUITS	4	9	A	
3	CONTROL SYSTEMS	4	9	A	
4	DIGITAL ELECTRONIC CIRCUITS	4	9	A	
5	SIGNALS & SYSTEMS	4	10	S	
6	ANALOG ELECTRONICS CIRCUITS LAB	2	9	A	
7	DIGITAL ELECTRONIC CIRCUITS LAB	2	9	A	
8	ENHANCING SELF COMPETENCE	2	9	A	
SGPA: 9.15	**CGPA: 9.47**				
IV SEMESTER					
1	PROBABILITY THEORY & NUMERICAL METHODS	4	10	S	May – June 2018
2	DIGITAL SIGNAL PROCESSING	4	10	S	
3	NETWORK ANALYSIS	4	10	S	
4	ELECTROMAGNETIC THEORY	4	9	A	
5	ANALOG COMMUNICATION	4	8	B	
6	DIGITAL SYSTEM DESIGN USING VERILOG	4	10	S	
7	DIGITAL SIGNAL PROCESSING LAB.	2	10	S	
8	DIGITAL SYSTEM DESIGN LAB.	2	10	S	
SGPA: 9.57	**CGPA: 9.50**				

Sl. No	Subjects	Credits	Grade Point	Grade Letter	Month & Year of Passing
V SEMESTER					
1	DIGITAL COMMUNICATION	4	8	B	Dec. 2018 – Jan. 2019
2	VLSI CIRCUITS	4	9	A	
3	LINEAR INTEGRATED CIRCUITS & APPLICATIONS	4	9	A	
4	TRANSMISSION LINES & WAVEGUIDES	4	9	A	
5	ADVANCED SIGNAL PROCESSING - ELE	3	9	A	
6	BASIC COMMUNICATION LAB.	2	9	A	
7	LIC & APPLICATION LAB.	2	9	A	
8	EMPLOYABILITY SKILL DEVELOPMENT - I	0	-	PP	
SGPA: 8.83	**CGPA: 9.38**				
VI SEMESTER					
1	ANTENNAS & WAVE PROPAGATION	4	10	S	May – June 2019
2	MICROWAVE DEVICES & COMMUNICATION	4	10	S	
3	SYSTEM DESIGN USING MICROPROCESSOR/MICROCONTROLLER	4	9	A	
4	INFORMATION THEORY & CODING	4	9	A	
5	EMBEDDED LINUX- ELE	3	9	A	
6	ANALOG AND MIXED MODE VLSI DESIGN - ELE	3	9	A	
7	ADVANCED COMMUNICATION LAB.	2	9	A	
8	SYSTEM DESIGN USING MICRO-PROCESSOR / CONTROLLER LAB.	2	9	A	
9	EMPLOYABILITY SKILL DEVELOPMENT - II	0	-	PP	
SGPA: 9.31	**CGPA: 9.37**				
VII SEMESTER					
1	EMBEDDED SYSTEMS	4	9	A	Dec. 2019– Jan. 2020
2	WIRELESS COMMUNICATION	4	9	A	
3	POWER ELECTRONICS	4	9	A	
4	LOW POWER VLSI– ELE	3	10	S	
5	PIC & ARM MICROCONTROLLER- ELE	3	10	S	
6	VLSI LAB	2	8	B	
7	SEMINAR	2	9	A	
8	PROJECT - PHASE I	1	10	S	
9	IMMERSIVE GROUP WORKSHOP	0	-	PP	
SGPA: 9.22	**CGPA: 9.35**				
VIII SEMESTER					
1	AUTOMATION USING SCRIPTING LANGUAGE – ELE	3	9	A	August 2020
2	ADVANCED DIGITAL LOGIC VERIFICATION - ELE	3	9	A	
3	COMPUTER COMMUNICATION NETWORKS	4	9	A	
4	POWER ELECTRONICS LAB.	2	9	A	
5	PROJECT – PHASE II	9	10	S	
6	INTERNSHIP	0	-	PP	
7	GRAPH THEORY - ELE	3	10	S	
SGPA: 9.50	**CGPA: 9.37**				

❖ **Cleared in Subsequent Exams.**

TOTAL CREDITS EARNED ARE **200** WITH A CUMULATIVE GRADE POINT AVERAGE OF **9.37**

PP: Passed the Mandatory Non Credit Course.

AUTHENTIC

CONTROLLER OF EXAMINATIONS

N.M.A.M. INSTITUTE OF TECHNOLOGY NITTE UDUPI 574 110

PRINCIPAL

Nitte, Karkala Taluk - 574 110,
Udupi District,
Karnataka, India

T: +91 8258 281 039 (D)
(EPBAX) 281 263, 281 461, 281 462, 281 248, 281 349
F: +91 8258 281 265

E: principal_nmamit@nitte.edu.in
W: www.nmamit.nitte.edu.in

Credit Ranges and Grading System

- **B.E. Degree is a 200 Credits programme and the major contents of this programme are**

	Credit Range
a) Basic Science Core courses	25 – 35
b) Engineering Science Core courses (Engg Foundation courses)	25 – 30
c) Humanities and Social Science Core courses	10 – 15
d) Programme Core courses	100 – 110
e) Elective courses	20 – 30
f) Departmental Program-Major Project	8 – 12
g) Mandatory Learning courses	Non-credits

- **2 years M.Tech Degree is a 100 Credits programme**
- **2 years M.B.A. Degree is a 100 Credits programme**
- **3 years M.C.A. Degree is a 150 Credits programme**

- Percentage equivalence of Cumulative Grade Point Average **(CGPA)** for the **TEN** Point Scale :

Grade Points	Percentage Marks & Class
5.00	42.5 Second Class (SC)
6.75	60 First Class (FC)
7.75	70 First Class Distinction (FCD)

Percentage = (GPA - 0.75)x10%

Performance of the students is evaluated continuously (CIE) and also by means of Semester End Examination (SEE), project work, viva-voce and other methods of testing as approved by the institution.

Grading System

Marks Scored	Letter Grade	Grade Point
≥ 90%	'S'	10
≥ 75% < 90%	'A'	09
≥ 60% < 75%	'B'	08
≥ 50% < 60%	'C'	07
≥ 45% < 50%	'D'	05
≥ 40% < 45%	'E'	04
< 40%	'F' (Fail grade)	00
	'N'	Not eligible
	'U'	Audit
	'W'	Withdrawal
	'PP'	Pass
	'NP'	Not Pass

SGPA for a semester is computed as follows :

$$\text{SGPA} = \frac{\sum[(\text{Course credit}) \times (\text{Grade point})] \text{ for all courses in that semester with Letter grades, including F}}{\sum[\text{Course credits}] \text{ for all courses in that semester with Letter grades including F}}$$

CGPA is computed as follows :

$$\text{CGPA} = \frac{\sum[(\text{Course credit}) \times (\text{Grade point})] \text{ for all courses with Letter grades except F, until that semester}}{\sum[\text{Course credit}] \text{ for all courses with Letter grades except F, until that semester}}$$

Urkunde

Albert-Ludwigs-Universität Freiburg

UNI FREIBURG

Technische Fakultät

Rahul Hegde

geboren am 1. Februar 1998

hat am 14. November 2023
die Masterprüfung im Fach

Embedded Systems Engineering

erfolgreich abgeschlossen.

Aufgrund dieser Prüfung wird der akademische Grad

Master of Science (M.Sc.)

verliehen.

Freiburg i. Br., 15. November 2023

F. Balle

Prof. Dr.-Ing. Frank Balle
Dekan der Technischen Fakultät

(Siegel)

EXA 256a (01/2023) / ID: 10306774

Zeugnis

Albert-Ludwigs-Universität Freiburg

UNI FREIBURG

Technische Fakultät

Rahul Hegde

geboren am 1. Februar 1998

hat am 14. November 2023 im Studiengang

Master of Science (M.Sc.)

die Masterprüfung im Fach

Embedded Systems Engineering

mit der Gesamtnote

gut (2,2)

bestanden und insgesamt 123 ECTS-Punkte erworben.

Masterarbeit: sehr gut (1,1)

Titel: 3D Mapping of Structured Environments Utilizing LiDAR Scanners and IMU Mounted on a Mobile Robot

Freiburg i. Br., 15. November 2023

Prof. Dr. Ulrich Egert

Prüfungsausschussvorsitzender

(Siegel)

Albert-Ludwigs-Universität Freiburg

Herr
Rahul Hegde
geb. am 01.02.1998

Leistungsübersicht/Transcript of Records

Studiengang: Master of Science (M.Sc.)
Hauptfach: Embedded Systems Engineering, PO 2021

	Status / Note	ECTS-Punkte	Semester / Bemerkung
ECTS-Punktekonto Master of Science in Embedded Systems Engineering (PO-Version 2021)		123	
Mastermodul / Master Module	1,1	30	
Masterarbeit 3D Mapping of Structured Environments Utilizing LiDAR Scanners and IMU Mounted on a Mobile Robot	1,1	27	WS 2023/24
Masterkolloquium	1,1	3	WS 2023/24
Informatik | Computer Science	2,9	42	
Essential Lectures in Computer Science	3,2	18	
Cyber-Physical Systems – Discrete Models	3,7	6	
Cyber-Physikalische Systeme - Diskrete Modelle / Cyber-Physical Systems – Discrete Models - Studienleistung	BE		WS 2020/21
Cyber-Physikalische Systeme – Diskrete Modelle / Cyber-Physical Systems – Discrete Models - Prüfung	3,7	6	WS 2020/21
Foundations of Artificial Intelligence	3,3	6	
Foundations of Artificial Intelligence - Prüfung: Grundlagen der Künstlichen Intelligenz/Foundations of Artifical Intelligence - Prüfung	3,3	6	WS 2021/22
Image Processing and Computer Graphics	2,7	6	
Bildverarbeitung und Computergraphik / Image Processing and Computer Graphics - Prüfung	2,7	6	SS 2022
Elective Courses in Computer Science	2,7	24	
Computer Vision	3,0	6	
Computer Vision - Prüfung	3,0	6	WS 2021/22
Introduction to Mobile Robotics	3,0	6	
Introduction to Mobile Robotics - Prüfung	3,0	6	SS 2021
Reinforcement Learning	3,3	6	
Reinforcement Learning - Prüfung	3,3	6	WS 2021/22
Robot Mapping	1,7	6	
Roboter-Kartierung / Robot Mapping - Prüfung	1,7	6	WS 2021/22
Mikrosystemtechnik	2,4	42	
Advanced Microsystems Engineering	2,6	24	
Assembly and packaging technology	1,7	6	
Assembly and packaging technology - Prüfung	1,7	6	Anerkennung Univ. FR
Micro-electronics	2,7	6	
Micro-electronics - Prüfung	2,7	6	Anerkennung Univ. FR
Modelling and System Identification	3,0	6	
Modelling and System Identification - Studienleistung	BE		WS 2020/21
Modelling and System Identification - Prüfung	3,0	6	Anerkennung Univ. FR
Sensors	3,0	6	
Sensors Lab Course - Studienleistung	BE		Anerkennung Univ. FR
Sensors - Prüfung	3,0	6	Anerkennung Univ. FR
Microsystems Engineering Concentrations Area	2,1	18	
Circuits and Systems		18	
Model Predictive Control and Reinforcement Learning	1,3	3	
Model Predictive Control and Reinforcement Learning - Studienleistung	BE		WS 2022/23
Model Predictive Control and Reinforcement Learning - Prüfung	1,3	3	WS 2022/23
Numerical Optimal Control in Science and Engineering	2,7	6	
Numerical Optimal Control in Science and Engineering - Studienleistung	BE		SS 2021
Numerical Optimal Control in Science and Engineering - Prüfung	2,7	6	SS 2022
Numerical Optimization	2,3	6	
Numerische Optimierung / Numerical Optimization - Studienleistung	BE		WS 2021/22
Numerische Optimierung / Numerical Optimization - Prüfung	2,3	6	SS 2022

	Status / Note	ECTS-Punkte	Semester / Bemerkung
Numerical Optimization Project	1,3	3	
Numerische Optimierung Projekt / Numerical Optimization Project - Studienleistung	BE		WS 2022/23
Numerische Optimierung / Numerical Optimization - Projekt	1,3	3	WS 2022/23
Materials and Fabrication	1,7	3	
Functional Safety, Security and Sustainability: Active Resilience	1,7	3	
Functional Safety, Security and Sustainability: Active Resilience - Studienleistung Übung	BE		Anerkennung Univ. FR
Functional Safety, Security and Sustainability: Active Resilience - Prüfung	1,7	3	Anerkennung Univ. FR
Customized Course Selection	BE	6	
Praktikum Informatik 1	BE	6	
Praktikum Informatik 1 (exam): Masterpraktikum Deep Learning (Robotics Track)	BE (1,3)	6	SS 2022

Master of Science in Embedded Systems Engineering **gut (2,2)** **123 ECTS-Punkte**

Datum der letzten Studien- bzw. Prüfungsleistung **14.11.2023**

Freiburg i. Br., 15. November 2023

Prof. Dr. Ulrich Egert
Vorsitzender des Prüfungsausschusses

Legende Status:
BE = Studien-/Prüfungsleistung bestanden; TRE = Teilnahme an der Lehrveranstaltung regelmäßig erfolgt

Legende Bemerkung:
Anerkennung Univ. FR = an der Universität Freiburg erbrachte Leistung; Anerkennung = nicht an der Universität Freiburg erbrachte Leistung.

Notenskala:
1,0 bis 1,5 = sehr gut
1,6 bis 2,5 = gut
2,6 bis 3,5 = befriedigend
3,6 bis 4,0 = ausreichend
5,0 = nicht ausreichend